

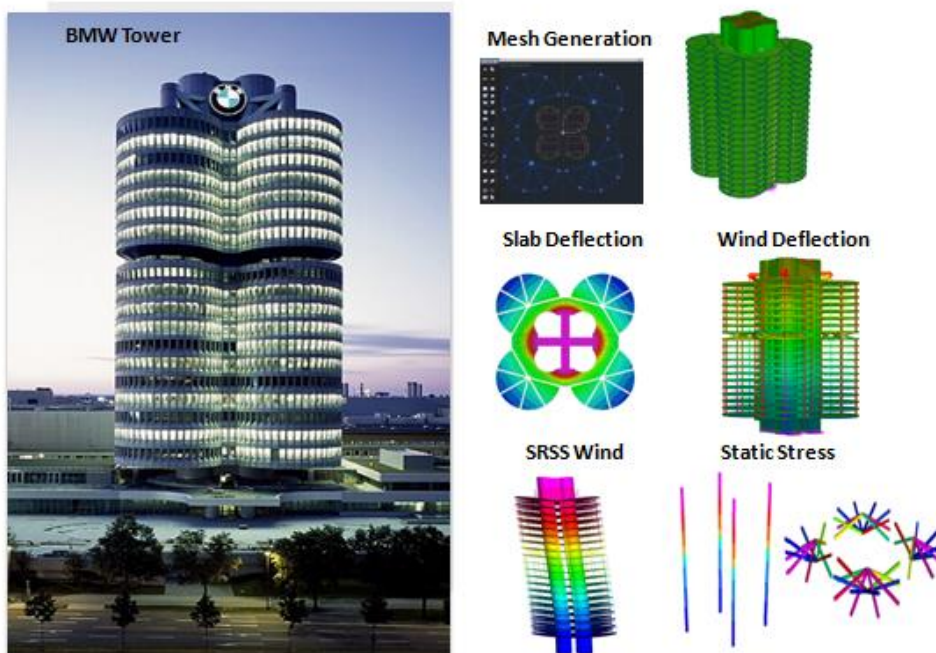
Dynamic Analysis of the BMW Tower in Munich

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Abstract. In the 1970s, world famous Austrian architect Karl Schwanzer designed an avant-garde suspended skyscraper for the new BMW headquarters. The BMW Tower was envisioned to resemble a four-cylinder motor and become a symbol for the recent flourishing success of BMW. Throughout its four decades, the BMW Tower has become the main architectural feature of modern Munich and a pride for one of the World leading car manufacturers. The structural design of the BMW Tower represented a major challenge to Germany's finest engineers because the suspended 99.5m-high structure had to withstand not only static loading but large wind dynamic loading while having deflections within appropriate serviceability limits. Strand7 has been used to determine the stresses and deflections the structure is subjected to in order to analyse its behavior under static and dynamic loadings. Ultimately, this analysis helps to understand the nature of suspended structures in relation to the Eurocode building standards. Finally, thermal resistance has also been analysed using Strand7 to simulate a fire scenario and analyse the behaviour of the cable structure, which is the most critical building component.



INTRODUCTION

The Austrian architect Karl Schwanzer, designed an award-winning multistory building for the BMW company. The design of the building was architecturally conceived to resemble the four-cylinder engine of a car. It is currently used as the headquarters for BMW.

The BMW Tower (also known as “BMW Vierzylinder”, “BMW Hochhouse”, “BMW Turm” in German) was constructed in Munich Germany between 1968 and 1972 with the engineer Helmut Bomard (Motorline, 2016). The use of suspended components makes the structure unique and iconic. Being next to a museum, the site attracts numerous international tourists to the region. This suspended structure is not only innovative but also opens numerous inspirational concepts that portray the various means of load bearing. The BMW Tower had been classified as a protected historical building since 1999 and it is a major landmark in Munich.

Being innovatively designed, the exterior has remained the same through time whilst the interior has been maintained in a futuristic manner in addition to office spaces and exhibition areas. At the shear central core, elevators and stairways are present- posing as the shortest routes to office spaces. ("BMW Four-Cylinder Building, Munich | LAMILUX Heinrich Strunz Group").

The building consists of four pre-stressed concrete cylindrical structures suspended off the ground through the use of cables connected to enormous cross-shaped steel hangers that lie on top of a shear core. With 18 suspended floors, a total of 53 000 m² of office space is available. These suspended floors consist of office space with one technical level mid building. This abundant space allowed the recentralization of staff previously dispersed around the region. A total land space of 72 000 m² was actually present, including the basement floor areas. Further details are given in Table 1.

This report presents the analyses conducted to understand the structure’s static and dynamic behaviors as well as the thermal resistance of the most critical structural components. Static behavior includes dead and live loads, while dynamic behavior comprises wind effects. Earthquake analysis was not part of this analysis as it is not required by the Eurocode for Munich because the probability of seismic activity is very low. (Solomos et al. 2008)

TABLE 1. Building Details

Location: Munich, Germany	The Year of Built: 1972
Architects: Karl Schwanzer	Approximate Cost: \$28.5 million AUD
Structural Engineer: Helmut Bomard	Overall Height: 99.5 m
Function: Commercial Office Building	Floor Area: 72000 m ² (52000 m ² useful area)
The Structure of the Plan: A circular floor plan with four segments representing a four-cylinder engine exists over 18 floors. The structure is suspended off hangers at the top of the building. The hangers are connected through the central shear core. There is also a technical floor mid-structure, containing trusses which transfer loads between the columns and the cables.	
Number of Floors: 18 Office Floors and 1 Technical Floor	

STRUCTURAL MEMBERS

Foundation: Raft slab for the shear core. Foundation over loose alluvial fine to coarse grained sediments that are the typical soil material in Munich, Germany.

Shear core: Four symmetrical reinforced-concrete cores along the entire height of the structure. Each core has a quadrant-like shape. Interior walls were used to stiffen the shear core. At each floor there is a stiffener slab used as corridor.

Hanger beams: Pre-stressed concrete cross-shaped hanger beams placed on top of the shear core structure.

- Cables:** Pre-stressed steel cable structure that are connected at the top to the hanger beam and on each floor is cast-in-place to a massive rebar display to assure a fixed connection. The cables have a concrete cover to protect them from fire.
- Columns:** Pre-stressed reinforced concrete columns at the edge of each radial cylindrical slab. Each column is at 7m spacing with floor-to-floor height of 4m. These columns are utilized to balanced out the loads in the steel cables. There are columns in all floors except for the technical floor.
- Beams:** Pre-stressed concrete beams extend radially from the floor-to-cable junction towards the exterior columns spanning 9m. Also, 7m peripheral beams are used to join the columns.
- Truss system:** Reinforced concrete 3-D truss system located on the technical floor that extends radially from the lower floor-cable junction to the base of the columns in the upper floor and from the top floor-cable junction to the top of the columns in the lower floor.
- Floor system:** Reinforced concrete two-way slab.

The members of the structure are determined from the original BMW tower. The use of material and dimensions of elements are modeled as per Table 2 below.

TABLE 2. Structural Elements for the BMW Tower

Details of the Structural Elements	Structural Element Sizes (mm)	Floor Level	Material and Standard
Hanger Beam (W x D)	1500 x 1500	Top floor only	Structural Steel – Eurocode 3
Cable (φ)	250	All	Structural Steel – Eurocode 3
Exterior Column (W x D)	200 x 200	All except technical	R. Concrete 40MPa – Eurocode 2
Floor Beam (W x D)	600 x 600	All	R. Concrete 40MPa – Eurocode 2
Truss (W x D)	600 x 300	Technical floor	R. Concrete 40MPa – Eurocode 2
Concrete Slab (D)	300	All	R. Concrete 40MPa – Eurocode 2
Outer Shear Wall (D)	350	All	R. Concrete 40MPa – Eurocode 2
Inner Shear Wall (D)	200	All	R. Concrete 40MPa – Eurocode 2
Stiffener Shear Wall (D)	175	All	R. Concrete 40MPa – Eurocode 2

STRUCTURAL SYSTEM

The structural system of this tower is built on the connections of elements. This system is modeled as per the sketches found in Fig. 1 and Fig. 2.

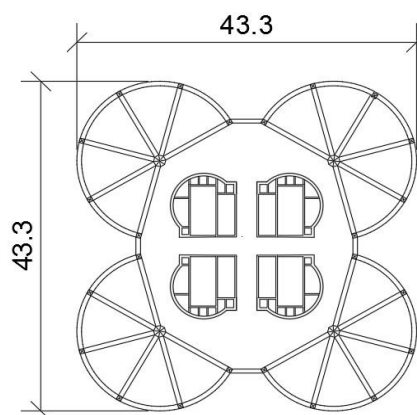


FIGURE 1. Plan of BMW Tower

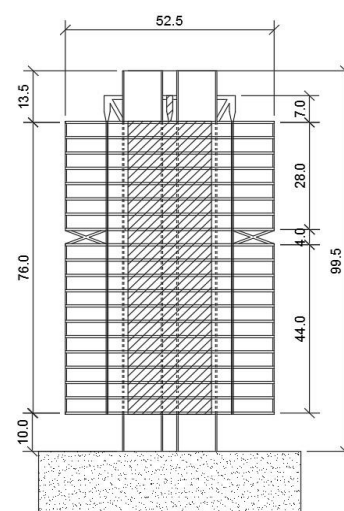


FIGURE 2. Side and Front Elevation of the BMW Tower

Construction

Initially, the concrete shear core was constructed to its full height. Steel hanger beams were then erected on top of the shear core. The steel hangers cantilever out from the shear core. The structures hang on cables that suspend from these steel hanger beams. The truss that comprises the technical floor was the first floor to be assembled. It was erected on the ground and is the most crucial structural system of the tower. All floors above the truss were then built and then the entire structure was raised one floor level by the steel cables. To construct each subsequent floor level situated beneath the truss technical floor, the floor slab was installed on-ground in the available space. The entire structure was then again elevated one floor level by the steel cables and the process was repeated. In the as-built state, the steel cables permanently suspend the structure off the ground. The truss provides a path for load redistribution between the cable and the exterior columns.

Vertical Loads

Vertical loading on this building is due to dead and live loads applied on the floor slabs. The floor slabs transfer these loads to the beams, from where it is transferred to the cables or exterior columns. The cables experience large tensile forces, which are redistributed into the exterior columns through the truss. The suspended system resists these forces by load transfer from the cables to the hangers and the shear core. The shear core is connected to the rigid foundation below therefore the suspended structure creates large compressive forces in the shear core.

Lateral Loads

Lateral loading would be produced by wind loading. While in reality, wind is applied to the exterior walls of the structure, in this model it is applied directly to the exterior columns. This lateral load is resisted by the transfer of forces from the columns through the truss, into the cables. These loads are then transferred into the hangers and shear core. In a dynamic response, the oscillating nature of the wind requires an efficient load transfer system, to reduce the warping that occurs from vibration of the structure.

Member Loads

Each member in the structure distributes load differently - depending on its property, orientation, and position.

Floor	The slab floor resists load by spreading both the dead and live loads across the floor into the beams and girders below. The rigidity of the slab floor also aids in the resistance in deflections- followed by the implementation of reinforcing bars.
Beams	Responsible for transmitting the live and dead loads to the columns and cable structure. Also, responsible for resisting the lateral loads due to dynamic wind effects and transfer such loads directly into the cable.
Columns	They transfer the floor and beam loads axially up through the structure. The columns connect through the truss to the cables, redistributing the tension and compression loads. Above the truss, the columns experience compression, and below the truss the columns experience tension.
Truss	Responsible for relieving the tensile forces in external columns by transferring loads to the cables.
Cables	The cables support the floor system of each storey. The cable can only experience tension and, therefore, all the loads on the structure are transferred through the cables to the hanger beams.
Hanger beams	Cross structure responsible for transferring loads from the cables to the shear core.
Shear core	Component that resists all the loads experienced by the structure. It experiences both axial loads imposed by the hanger beams and lateral loads imposed by wind effects on the very shear walls.

LOADS

Static Loading

In Strand7, the dead load was applied through a structural mass load case, with gravity of -9.81m/s^2 applied to the structure. The dead load on the building can be evaluated through the summary function on Strand7. The entire

structure consists of a mass of 3.52×10^7 kg. This mass averaged between the total slab areas, produces a load of 15.71 kPa acting on each slab.

$$G = 15.71 \text{ kPa} \quad (1)$$

Live loads are determined as per BS-EN-1991-1-1:2002 as 3 kPa for an office building. This force is applied on all floor slabs, except for the roof slab, as a distributed load. The Snow load was also determined from BS-EN-1991-1-1:2002 as 0.4 kPa for Munich.

$$Q = 3 \text{ kPa} \quad (2)$$

Wind actions were determined from BS-EN-1991-1-4:2005 with a base wind velocity of $v_b = 25$ m/s for Munich and an average air density of 1.25 kg/m^3 . The calculation considers the following equations to determine the wind force dependent upon the height being considered, $q_p(z)$, using curve IV as shown in Fig. 3 to calculate $c_e(z)$.

$$q_p(z) = c_e(z) \cdot q_b \quad (3)$$

$$q_b = 0.5 \cdot \rho \cdot v_b^2 \quad (4)$$

Because $c_e(z)$ was provided as a graph in BS-EN-1991-1-4:2005, the function was reproduced by graphing points into excel with an exponential trendline. This enabled the wind load to be applied as a function of the height of the structure on each column in Strand7 (Fig. 4). The equation used was:

$$c_e(z) = 0.7622 \ln(z) - 0.617 \quad (5)$$

Since the wind load was applied onto the exterior columns on one side of the structure, q_b was multiplied by the tributary width of each column. The spacing between subsequent columns differs across each half of the structure. Therefore, the loading width was taken as the sum of half the distance between columns on either side.

For the ultimate strength limit state design according to Eurocode 2 (BS EN 1992-1-1:2004), the following combination will be used for permanent, imposed and wind loads:

$$1.35 G + 1.35 Q + 1.35 W \quad (6)$$

Dynamic Wind Loading.

To determine the response of the structure to a dynamic wind loading, a spectral response was conducted with modes determined from the natural frequency analysis. Dynamic wind curves are usually derived from wind gusts in the surrounding location with a Fourier Transform to transform the gust signal into a frequency curve (Strand7 Webnotes, 2015).

The governing equation of the spectral response analysis is (Alonso-Marroquin, 2016):

$$M\ddot{U} + C\dot{U} + KU = F(t) \quad (7)$$

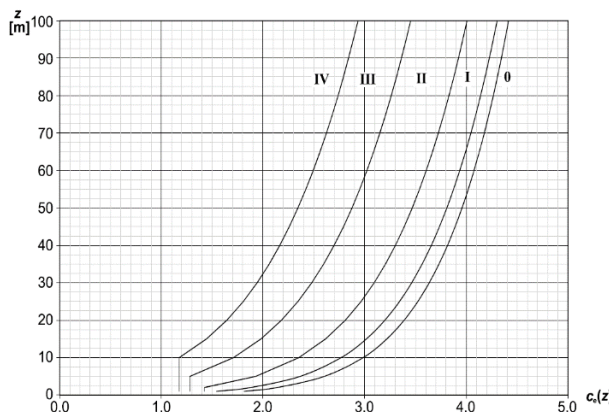


FIGURE 3. $c_e(z)$ factor for Static Wind Loading

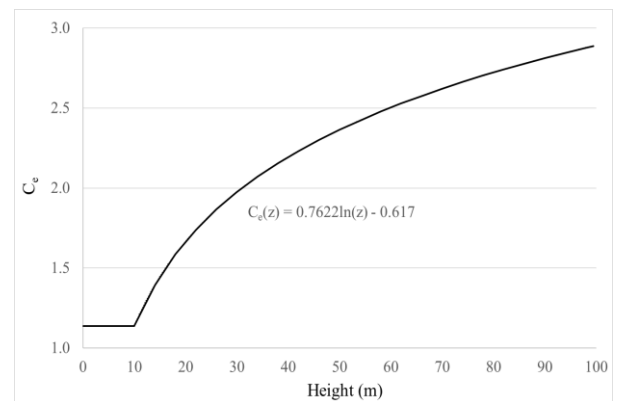


FIGURE 4. Wind load factor as a function of height

Where U is the structure displacement matrix, M is the mass matrix, C is the damping matrix, K is the stiffness matrix, and $F(t)$ is the matrix of the externally applied forces acting on the structure.

The displacement, U , is dependent upon the vibration frequencies, ω , as shown in Eq. 8 below (Alonso-Marroquin, 2016):

$$U(t) = U_0 e^{j\omega t} \quad (8)$$

Hence, in order to conduct this dynamic wind analysis, the structure's vibrational modes must be determined to find the response of the structure. The spectral response will depend upon the combination of the vibrational modes effect on the structure, and the associated load excitation for each mode.

The vibrational modes of a structure can be found by determining the natural frequencies through a Natural Frequency Analysis. This analysis removes the effect of the acting damping and external forces from the structure, resulting in the equation of the unforced-undamped vibrational motion of a multidegree of freedom (MDOF) system:

$$M\ddot{U} + KU = 0 \quad (9)$$

From Eq. 9 the natural frequencies, ω , can be determined as eigenvalues, and the mode shapes can be calculated as their corresponding eigenvectors.

The spectral response can then be determined by applying a static wind load to the structure. The static wind load is the externally applied force, $F(t)$, and was calculated as previously described in the Static Loading Section. In the Strand7 spectral response solver, this corresponds to choosing the applied load option.

The effect of the static wind loading was combined with the spectral response curve to simulate the displacement response, U , to dynamic loading experienced by the BMW Tower. It should be noted that this structure has attempted to simulate the discontinuous shear core due to elevator door openings.

A Power Spectral Density (PSD) curve is used to determine the dynamic spectral response. A PSD curve represents a typical wind gust curve. According to Davenport's approach (1961) it is assumed that "although gusts occur in a purely random sequence their statistical properties over a suitable period can be regarded as constant and independent of the origin of time." This period of time ranges from 5mins to 1 hour. Davenport determined the typical normalised PSD for along wind, which is dependent on the base wind, v_b , of the site. It produces a normalised factor vs frequency curve as follows (Cheimets, 1990).

$$Factor = 4.0 \frac{\left(1200 \frac{n}{V}\right)^2}{\left(1 + \left(1200 \frac{n}{V}\right)^2\right)^{\frac{4}{3}}} \quad (10)$$

Where, n is frequency in Hertz and V is average velocity, which is taken as 25 m/s for Munich.

The Davenport PSD is as follows in Fig. 5 below. For a base wind velocity of 25 m/s, the normalised PSD peaks at a frequency of 0.036 Hz corresponding to a period of 27.8 s, which is expected from wind effects on buildings. This a relatively low frequency compared to peak frequencies from Earthquake loading. According to Davenport (1961), "The effect of increasing the wind speed would be to increase the intensity of the fluctuations and to sweep the spatial pattern past a point on the ground faster, but without actually altering the pattern."

In Strand7, a natural frequency analysis is conducted first to determine the vibration modes of the structure. Once the modes are determined, a Spectral Response analysis is conducted to determine the response of the structure to the applied static wind load. The normalised Davenport PSD curve is used in conjunction with the static wind load to determine the actual dynamic response of the structure due to wind loading in the direction of the main wind pressure. Additionally, 5% damping was applied to the structure.

Thermal Loading

The thermal load has been applied only to the pre-stressed steel cable of the structure. Only the cable has been considered because it is a critical feature of the structure. This is because the structure is a low-redundancy system and if the steel cable yields, it would reduce in strength and cause failure of the building. Hence, it is important to

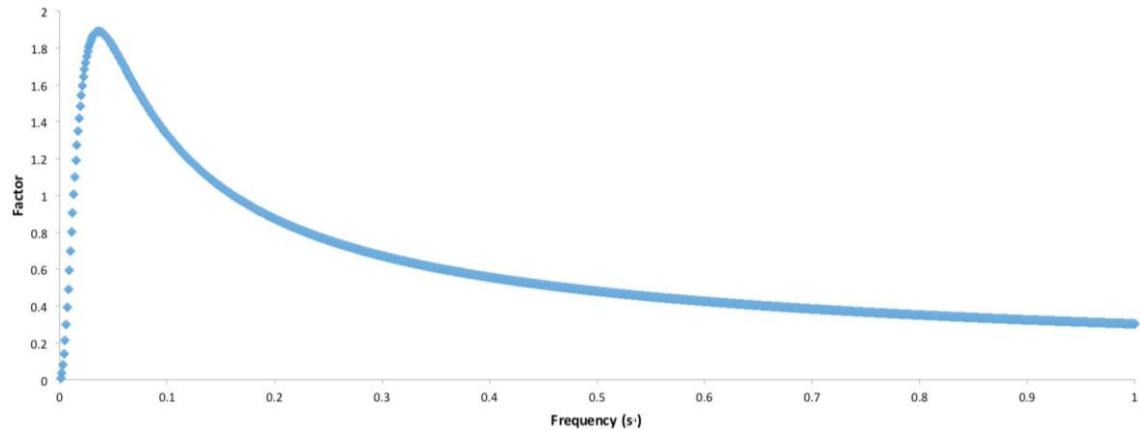


FIGURE 5. Davenport PSD Curve

ensure there is sufficient time for evacuation. The thermal load has been determined from BS-EN-1992-1-2:2004 as a maximum temperature of 1200°C for fire loading. These cables are enclosed within a thick concrete layer, which provided the thermal resistance in the steel.

In Strand7, a transient heat analysis will be conducted with calculations completed at time intervals of 15 min. An initial temperature of 25°C ambient temperature was applied to the structure to simulate the typical internal temperature of the building. The outer boundary conditions do not allow heat transfer. The heat conducts between the concrete and steel. The steel core was included to increase accuracy of the results. This accounts for the conductive effects of the steel core.

NUMERICAL ANALYSIS

Mesh Generation

A blueprint of the BMW Tower floor plan was found from recent renovations. This file was placed into AutoCAD, from where measurements were taken to determine sizing of beams, columns, floor slab and shear core. The completed AutoCAD section is shown in Fig. 1.

Once this was completed, the AutoCAD .dxf file set up was imported into Strand7. Different layers in AutoCAD (shown by different colours in Fig. 6) are automatically generated as separate groups in Strand7.

In Strand7, polylines were recognised as beams, while closed polylines were recognized as plates. Therefore, the beams, cable and column connecting plates were instantly created. From there, the column and cable plates centre nodes were extruded to become truss elements. The shear wall beam elements were extruded to become plates and formed the vertical shear wall.

To form the floor slab, the triangular sections between beams were subdivided manually to ensure the correct subdivision would produce a constant mesh across the floor slab. Once the floor slab was set-up, this was incrementally copied up the structure.

In the technical floor, the columns were deleted while beam elements were used to create the radial 3D-truss system. Also,

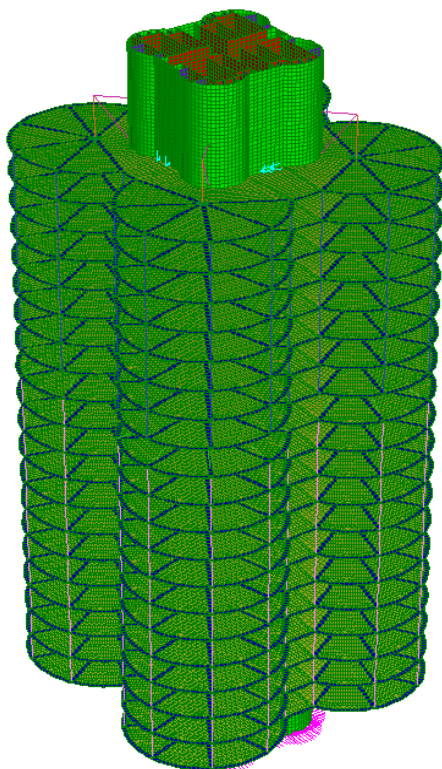


FIGURE 6. Modeled Structure

on the upper part of the shear core, beam elements were used to create the cross-shaped hanger structure.

Because the slabs were lifted from the ground, they have been modeled with nodes disconnected from the shear core. Finally, links were used on each floor to partially connect the slab to the shear core to simulate artificial connections that might exist in the real structure.

During the set-up of the Strand7 model, the connection between elements has been thoroughly done in order to achieve continuous results except in the case of the connection between the slab and the shear core, as previously explained. Table 3 presents a detailed summary of the elements used to model the structure. The material geometric properties were shown in Table 2. Finally, Tri3 and Quad4 plate elements were chosen in order to alleviate the computational time to run each model. However, more accurate results could be achieved with Tri6 and Quad 8 plate elements.

TABLE 3. Strand7 Elements used to Model the BMW Tower

Details of the Modelled Elements	Material	Strand7 Element	Number of Elements
Hanger Beam (W x D)	Structural Steel	Beam	24
Cable (ϕ)	Structural Steel	Truss	80
Cable Plate (W x D)	R. Concrete 40MPa	Plate/Shell Tri 3	640
Exterior Column (W x D)	R. Concrete 40MPa	Truss	432
Column Plate (D)	R. Concrete 40MPa	Plate/Shell Tri 3	1920
Floor Beam (W x D)	R. Concrete 40MPa	Plate/Shell Tri 3 & Quad 4	11040
Truss (W x D)	R. Concrete 40MPa	Beam	96
Slab (D)	R. Concrete 40MPa	Plate/Shell Quad 4	85200
Outer Shear Wall (D)	R. Concrete 40MPa	Plate/Shell Quad 4	26725
Inner Shear Wall (D)	R. Concrete 40MPa	Plate/Shell Quad 4	46204
Stiffener Shear Wall (D)	R. Concrete 40MPa	Plate/Shell Quad 4	11940

Solvers

A detailed list of solvers for each analysis is shown in Table 4.

TABLE 4. Solvers used to model the BMW Tower

Load Type	Static	Dynamic Wind	Thermal
Solver	Linear Static	Natural Frequency Spectral Response	Transient Heat
Input	Elastic Modulus Poisson's Ratio Density	Density Vibrational Modes Factor Vs. Frequency Table	Initial Temperature Specific Heat Thermal Conductivity
Output	Displacement Stresses Permanent	Displacement Stresses	Temperature as a function of time
Load	Imposed (live and snow)	Permanent Imposed (wind)	Fixed Temperature
Boundary Conditions	Fixed at Base	Fixed at Base	Fixed external nodal temperature of 1200°C

Static Deflection Results

The load combination effects on the structure under serviceability limits were analysed with an applied load combination of $G + Q + W$ (dead, live and wind). Fig. 7 visualises the DY deflections under this serviceability state limit. The majority of these deflections are due to the applied lateral loading on the windward and leeward sides in the direction of y . The dead and live loads do not generate any lateral deflections; however a second order analysis could be done to resemble more realistic results. The structure as a whole experiences a maximum of 37mm deflecting laterally in both beams and plate elements. This maximum deflection is located at the connection between the angled hanger and the shear wall, hence representing both plate and beam elements. This point is

where the loading is transferred directly into the shear wall from the loaded columns and beams. Therefore, maximum loading causes the highest displacement in the structure.

Since the wind loads are increasing with height and are applied at the external columns, it is expected that there is a greater deflection at the top of the structure relative to the base. This deflection is evident in both the external columns and shear core as per Fig. 7. The linear deflections decrease down the vertical plane.

Vertical deflections in the DZ direction are analysed in Fig. 8. The range of plate deflections experienced within the structure varies from 7mm to 260mm. All the beam elements deflect within this range of values. Considering that beam elements are representing both the hangers and the steel cable-deflections, they will experience the same deflection as the plates at those points. The slightly larger deflections are purely due to the cantilevered extension of the slabs from the cable connection. From Fig. 8, it is also clear that the shear core experiences a uniform minimum deflection down the entire structure, as the loading transferred through the hangers affect the entire structure evenly.

It can be viewed more extensively through Fig. 9 and Fig. 10 which show deflections at the top and bottom slabs. Both images show the deflections increasing symmetrically towards the outer perimeter of the slab. This is due to the sagging experienced from the lack of support of a suspended structure. The support acts solely at the cables, and sagging will occur from this point out. It is also important to point out that the bottom slab experiences the highest deflections in the structure. This is because it is located at the bottom of the cables- which is holding up the entire structure's dead and live loads. The significant tension experienced from the cables are reflected in this large displacement. Comparing top and bottom floor slabs- there is a 251mm deflection at the bottom floor compared to 218mm at the top floor. This difference averages out to be approximately 16.5mm increase per floor due to the weight above.

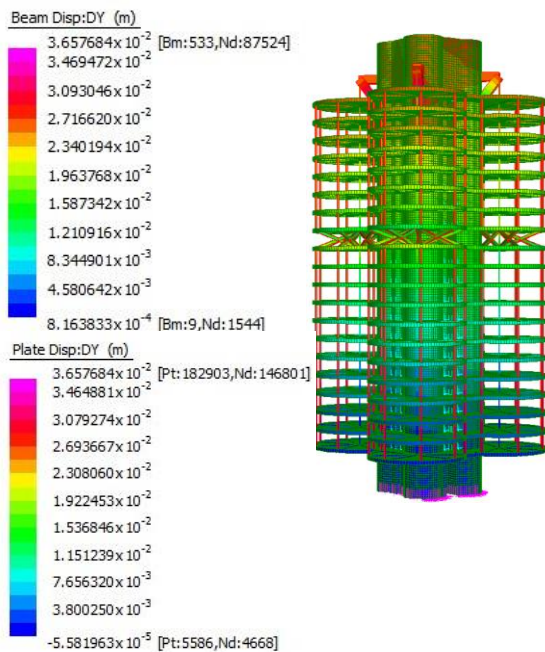


FIGURE 7. Deflection DY under Serviceability State Limit

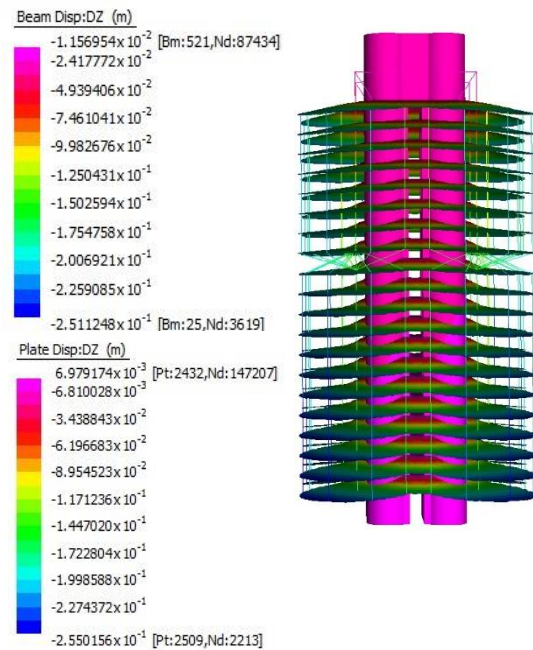


FIGURE 8. Deflection DZ under Serviceability State Limit

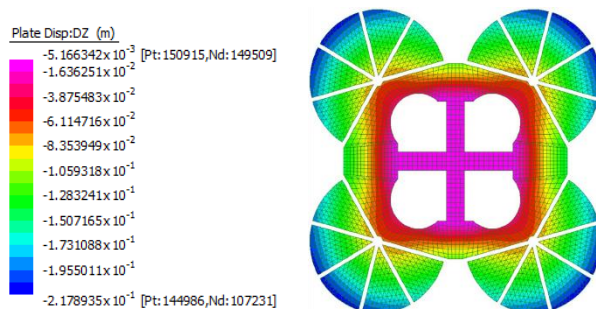


FIGURE 9. Slab Deflection at Top Floor

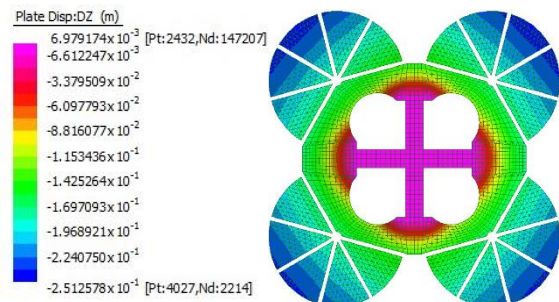


FIGURE 10. Slab Deflection at Bottom Floor

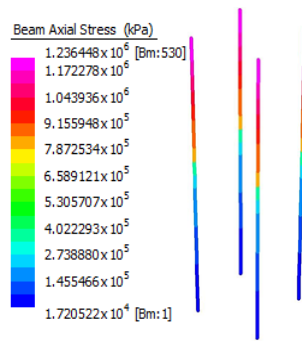


FIGURE 11. Axial stresses on cables

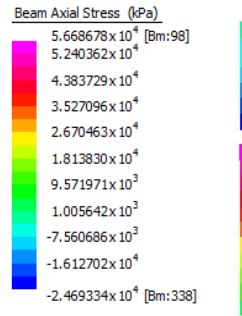


FIGURE 12. Axial stresses on a single column

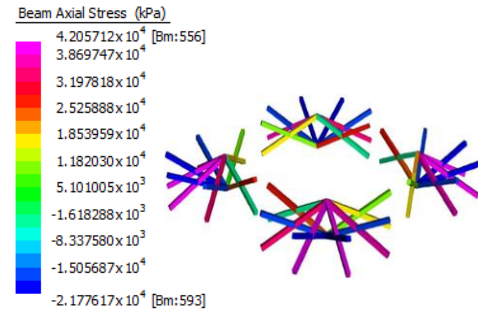


FIGURE 13. Axial stresses on Truss

Static Stress Results

The stresses are analysed for the cable, column, and truss elements of the structure as these will experience the largest stresses. These results are taking the appropriate combination for USL. Results will help further explain the transfer of loads. The cables experience the greatest axial tension in the structure due to the suspension of the slabs as shown in Fig. 11. This tension reduces further down the structure, to a minimum of 17.2MPa. The columns analysed are exterior columns at the perimeter of the slabs. It is clear that they are under compression at the upper section of the building (above the truss level) and under tension at the lower section of the building (below the truss level) as shown in Fig. 12. The maximum compression and tension values are both at the connecting points through the truss level. Therefore, it is evident to conclude that the truss functions to redistribute the loads through the columns from the cables. In the upper section, the tension in the cables is balanced out through the compression in the column, dissipating in axial stress at the top. Similarly, for the lower columns, the tensile stresses experienced dissipate away from the truss level. The truss level clearly shows a combination of tensile and compressive stresses as per Fig. 13. They successfully redistribute loads through these members, connecting through to the external column and internal steel cable, as described earlier. Evidently, the compressive truss members are linking the compressive external columns to the cable whilst the tensile truss members are linking the lower tensile columns to the cable. The inner truss members are seen to further redistribute the loads in various directions.

Dynamic Wind Deflection Results

Initially, a natural frequency analysis was conducted 25 modes converged, as shown in Table 5.

TABLE 5. Natural frequencies of the BMW Tower

Mode	Frequency (Hz)	Mode	Frequency (Hz)	Mode	Frequency (Hz)
1	0.500	9	2.440	17	3.460
2	0.531	10	2.470	18	3.480
3	0.885	11	2.600	19	3.500
4	1.160	12	2.790	20	3.510
5	1.170	13	2.850	21	3.530
6	1.340	14	3.260	22	3.740
7	1.390	15	3.310	23	3.760
8	2.240	16	3.440	24	3.820
				25	3.890

After the convergent modes were determined, a Spectral Response Analysis was conducted to investigate the effects of dynamic wind on the structure. The static wind load was applied to the structure, with the Davenport PSD as described previously in Dynamic Wind Loading..

The dynamic wind analysis determined that mode 5 was the most significant mode, with a maximum excitation amplitude of 8.82 produced. This was much greater than the other modes and the SRSS combination can be considered to represent this mode.

The dynamic wind analysis produced the deflected shape as shown in Fig. 14, at 10% exaggerated deflection. The displacement of the shear core at the height of the building is 13 cm in the along wind direction (DY). This is a relatively small deflection, but is still four times larger than the deflection determined by the static analysis, 2.8cm as shown in Fig. 7. The increase in the dynamic analysis is expected because the analysis responds to a peak wind loading from a typical wind gust curve that the PSD represents. Additionally, that these results are in a similar frame of size shows that the analysis was effective in determining the deflections.

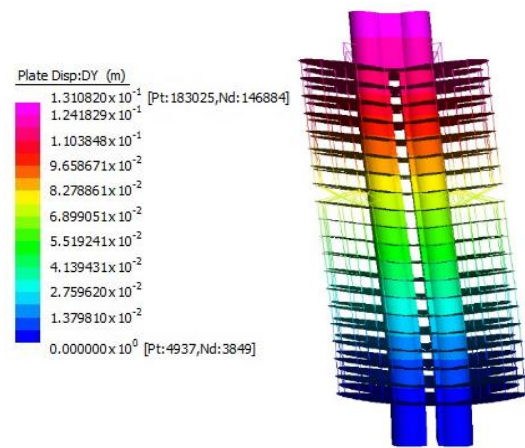


FIGURE 14. Displacement due to Dynamic Loading

Resistance to Thermal Effects

This analysis was performed through a two-dimensional model in Plain Strain, using Quad8 plate elements to model both the concrete cover and the steel cables inside. The cables had a total diameter of 250mm, and the concrete cover to this steel was 250mm thick. Fig 15 displays the Strand7 model.

Through a transient heat analysis, the resulting heat transfer through the structural element can be understood. The maximum fire loading, according to the most conservative Eurocode guidelines was 1200°C. This load was applied to the external nodes to represent a highly uncontrolled fire, whilst other nodes were subjected to an initial 25°C ambient temperature.

If considering the failure of such an important element, the entire suspended floors will collapse as a result due to the amount of load taken by these cables. Therefore, it is critical to ensure that the cables have sufficient fire protection.

As shown in Fig. 16, the prestressed concrete encased cable was subjected to different heat transfer scenarios dependant on the heating times. It is apparent that there is a linear increase in temperature through the concrete cover towards the steel cable at any one time. However, temperatures in the cable are approximately constant throughout it's surface due to the elevated conductivity of steel. By including the steel cable inside the concrete cover, appropriate conduction of heat can be analysed through this cover, promoting a more accurate indication of fire resistance period (FRP). Considering a critical temperature of steel as 550°C (i.e. temperature at which the steel yield strength has been reduced to 60% of the initial yield strength), the present cable will have a FRP of up to 615min, which is equivalent to 10.25 hours. This time period is definitely sufficient for the safe evacuation of all office staff during a fire. Since the fire considered was of the worst type, generally there would be a greater time period given a smaller temperature. Regardless, it can be concluded that the cover is satisfactory for this multistorey building.

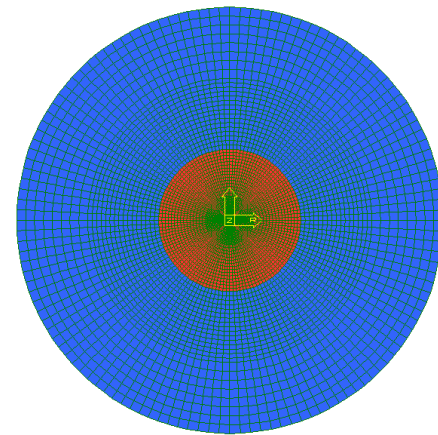


FIGURE 15. Model of Cable. Reinforced Concrete surrounds inner Steel Cable

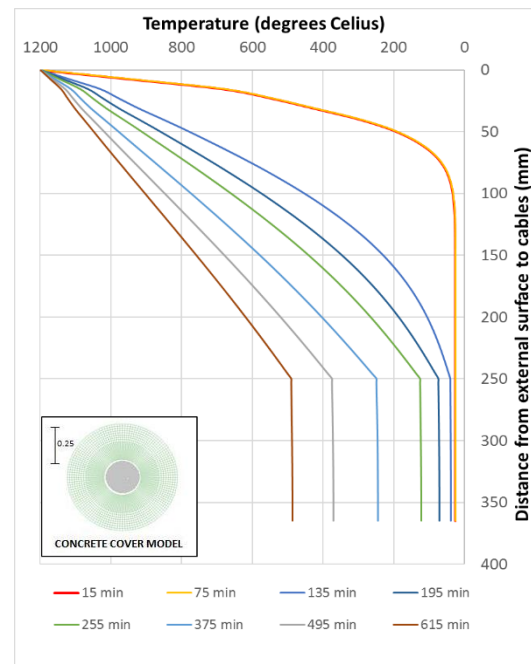


FIGURE 16. Heat dissipation through the cable's protective cover

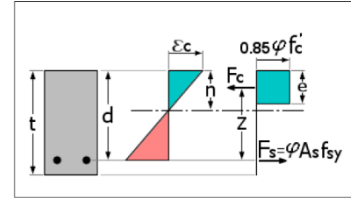
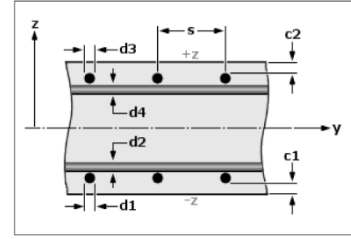
STRUCTURAL DESIGN

The initial model set-up did not consider reinforced concrete, but only plain concrete. Then, the analyses performed with Strand7 yielded the design effects on each of the structural members so it was actually possible to design each member for both ultimate strength level and serviceability strength level criteria.

Hence, a new model was run considering a first set of reinforcement layouts. This process was repeated iteratively until a solution that could satisfy the SSL criteria was determined. Table 6 shows summarizes the final rebar layouts for each type of structural element:

TABLE 6. Summary table for the reinforcement used

		Beams	Floor Slabs	Shear Walls
Cover	C1 (mm)	25	25	50
	C2 (mm)	25	25	50
Bar diameters	D1 (mm)	50	20	80
	D2 (mm)	50	20	-
	D3 (mm)	50	20	80
	D4 (mm)	50	20	-
Bar spacings	S1 (mm)	80	150	100
	S2 (mm)	80	150	-
	S3 (mm)	80	150	100
	S4 (mm)	80	150	-
Concrete properties	E _c (MPa)	32,800	32,800	32,800
	ε	0.003	0.003	0.003
	F' _c (MPa)	50	40	40
	φ	0.8	0.8	0.8
Rebar properties	E _s (MPa)	200,000	200,000	200,000
	F _{Sy} (MPa)	500	500	500
	A _{s_b}	0.5 t	0.5 t	0.5 t
	(base)			



Special consideration was given to the structural design of cables. Initially, the cables were estimated to have the following characteristics:

Axial Force:	60690kN
Diameter:	160mm
Cross sectional Area:	20106 mm ²

To verify this, GALFAN ® Coated Steel-Fully Locked Strands, PFEIFER, were used in the following calculations, where for a single strand:

Minimum Breaking Load (MBL)	405kN
Nominal Diameter	21.0mm
Cross-section Area	281.0mm ²

Currently a 160mm diameter cable contains 71 strands, with each having a minimum breaking load of 405kN.

$$N_{design} = n_{cables} \cdot MBL \quad (11)$$

$$N_{design} = 71 \cdot 405 \text{ kN} \quad (12)$$

$$\therefore N_{design} = 28760 \text{ kN} \leq 60690 \text{ kN} \quad (13)$$

Therefore a design axial force of 28760kN does not satisfy requirements of our current analysis. Giving a failure allowance factor of $\phi = 0.85$, the cable must be able to support approximately 71400kN. From this the entire cable is increased to a diameter of 250mm in the model.

$$N^* \leq 0.85 n_{cables} \cdot 405 \text{ kN} \quad (14)$$

$$\therefore 405 \cdot n_{cables} \geq \frac{60690}{0.85} = 71400kN \quad (15)$$

$$\therefore n_{cables} \geq \frac{71400}{405} = 176 \text{ cables} \quad (16)$$

$$176 \cdot 281 = 49456 \text{ mm}^2 \quad (17)$$

$$D = \sqrt{4 \cdot \frac{area}{\pi}} = 250mm \quad (18)$$

CONCLUSIONS

This paper provides a complete dynamic analysis of the BMW Tower, situated in Munich. It provides an investigation into static loading, dynamic wind loading, and thermal loading of the steel cable.

There is little technical information available regarding the structural behaviour of the BMW Tower. Consequently it has been necessary to perform an iterative process in order to be able to determine the structural design of the main components of the building structure.

The 3D finite element model, used in conjunction to the Eurocodes has been successful in estimating the sizing of the members. Initially, the model was run to determine the design effects on each structural element. Afterwards, the structural members were progressively redesigned to comply ultimate strength levels and serviceability strength level criteria. The finite element model of the BMW Tower considered the appropriate combination for static loads (dead and live) as well as the governing wind dynamic loading case for tall buildings. Therefore the structure was determined to have satisfactory serviceability deflections, and satisfactory USL in the cable.

Further research and optimization should be performed to achieve more accurate results. These may include, but are not limited to: using higher order elements such as Tri 6 and Quad 8, analysing the settlement effect of the soil foundation due to the structure, considering the effects of across-wind on the dynamic behavior of the building, evaluating the potential effect of a blast near the cable structure and recreating the design of the structural members using the German building code that was current in the 1970s.

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